2024 Lecture Notes

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2024-01-29

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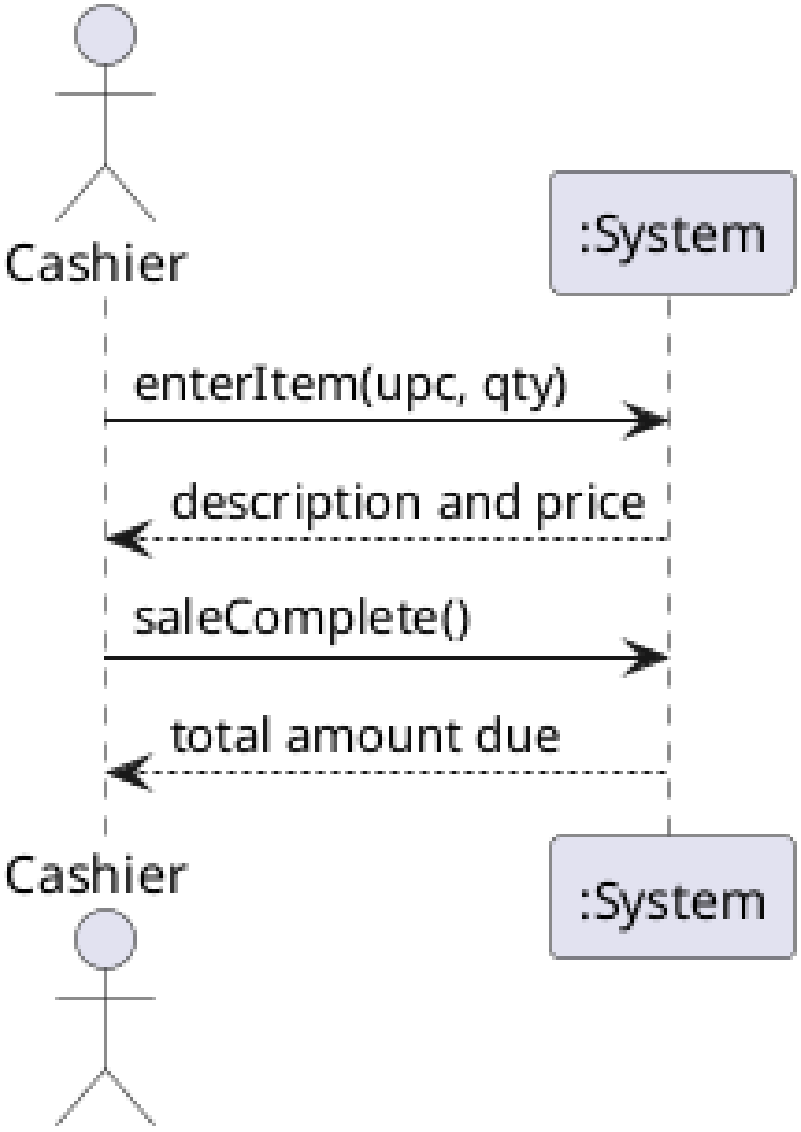
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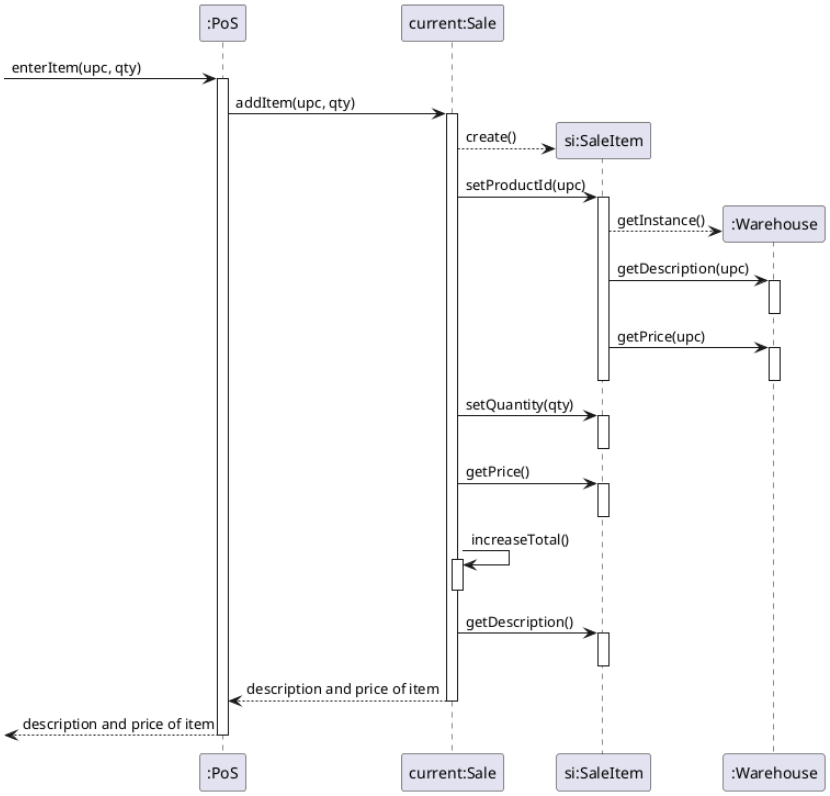
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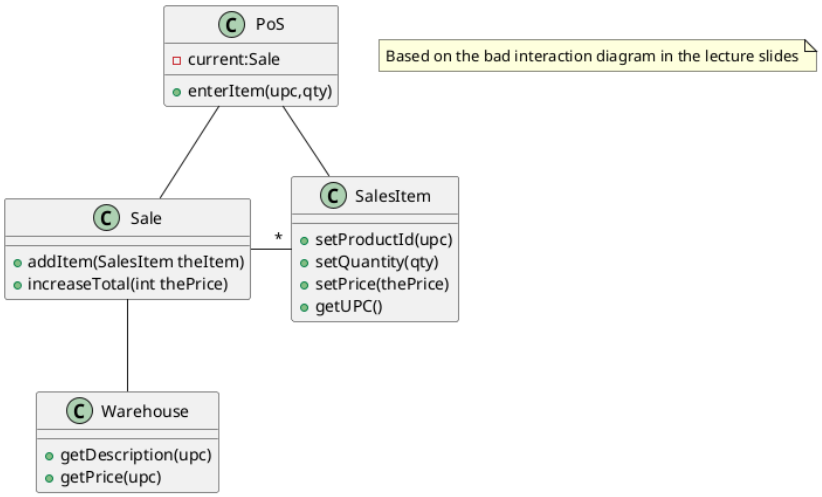
1 System Sequence Diagram



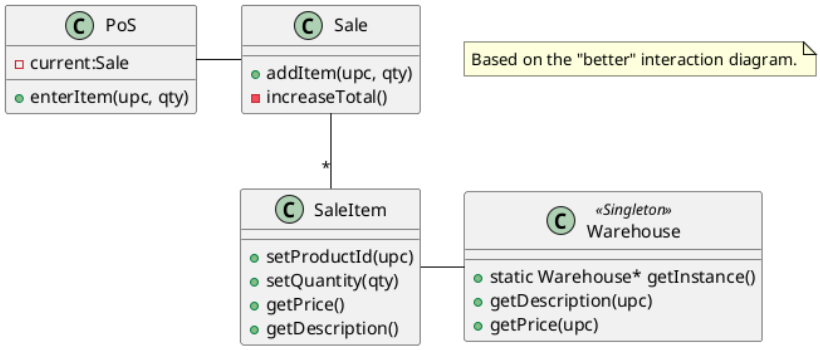
2 Interaction Diagram enterItem()



3 Class Diagram from Bad Interaction Diagram



4 "Better" Class Diagram



5 BurgerOrderer

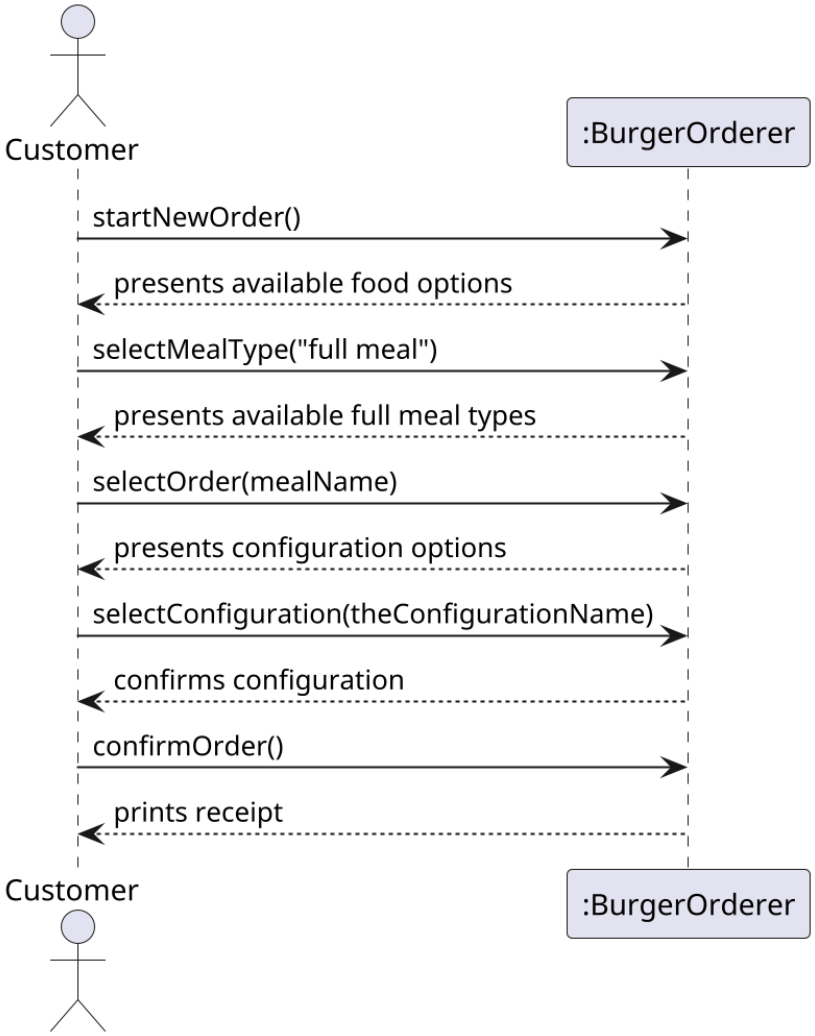
5.1 Use Case: Order Food

Use Case Order Food
Actors Customer
Description Customer orders a meal and configures it. The system registers the order and confirms.

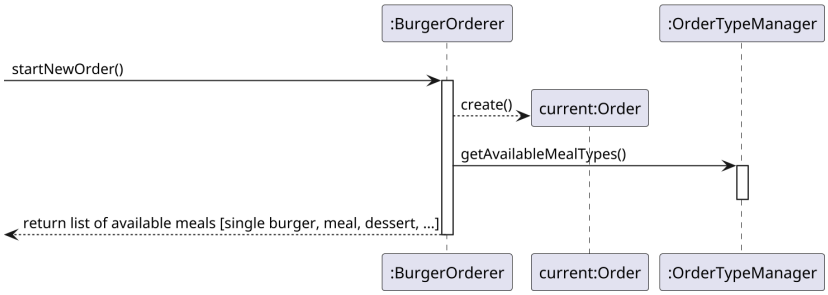
Main Course of Events

Actor	System
1. Customer arrives at BurgerOrderer and starts a new order.	
	2. System presents options [single burger, meal, dessert, . . .]
3. Customer selects "full meal"	
	4. System presents available meals
5. Customer selects a meal	
	6. System adds the selected meal to the o and presents configuration options.
7. Customer selects "no onions"	
	8. System adds "No onions" to the order
9. Customer selects "more bacon!"	
	10. System adds "more bacon!" to the ord
11. Customer confirms order	
	12. System initiates the use case pay for c
	13. System places burger order to tke Kit
	14. System prints receipt.

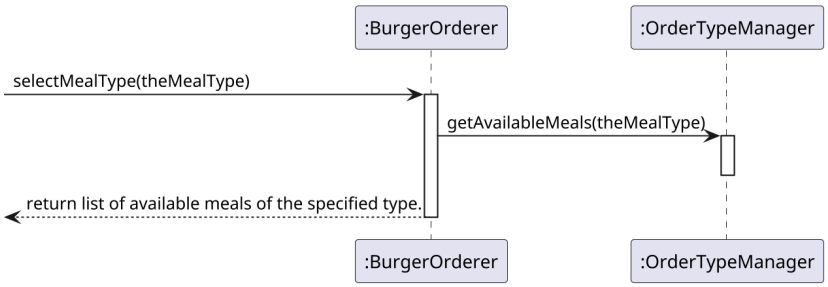
5.2 System Sequence Diagram



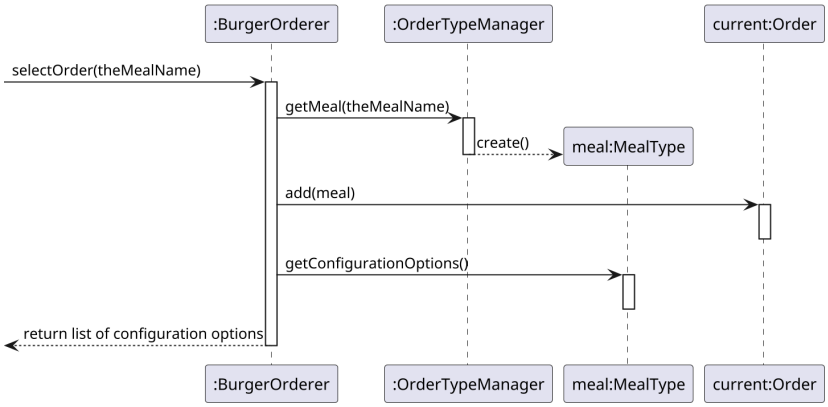
5.3 Sequence Diagram: startNewOrder()



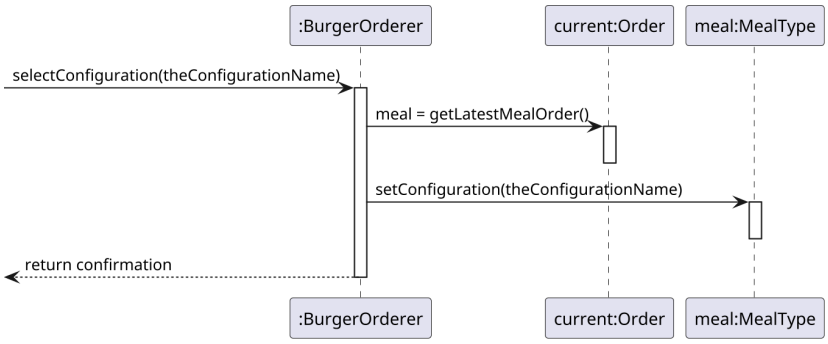
5.4 Sequence Diagram: selectMealType()



5.5 Sequence Diagram: selectOrder(theMealName)

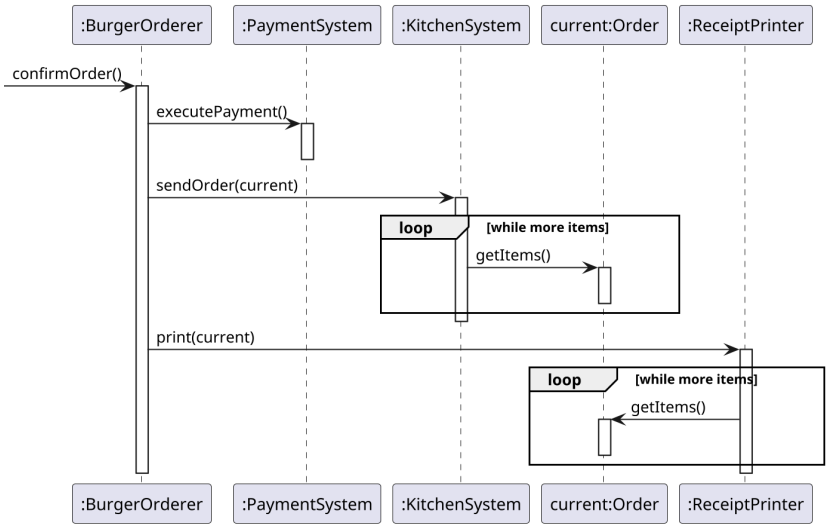


5.6 Sequence Diagram: selectConfiguration(theConfigurationName)

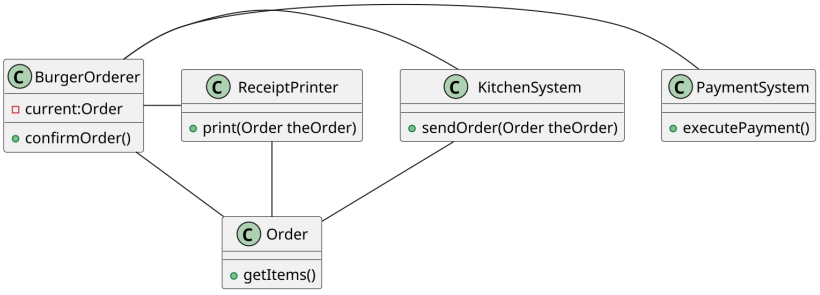


5.7 Sequence Diagram: confirmOrder()

12. System initiates the use case pay for order
13. System places burger order to tke Kitchen
14. System prints receipt.



5.8 Class Diagram

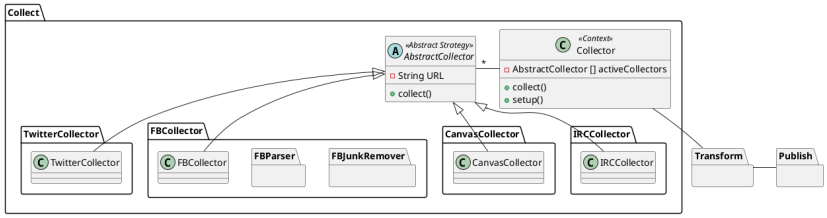


6 Example: SMSprucerUpper

6.1 Conceptual Model



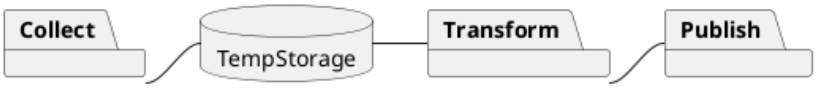
6.2 Package Model : Deeper into the Collect Package



6.3 The collect() method

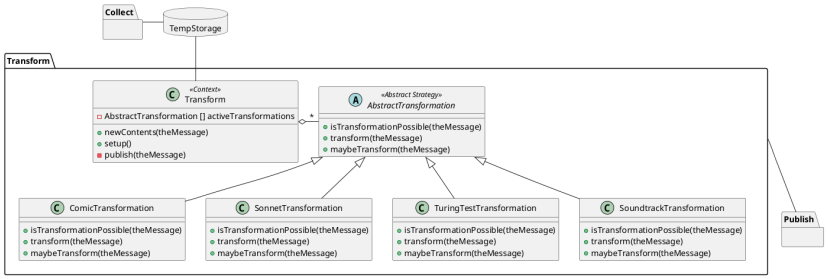
```
void Collector::collect() {
  for (AbstractCollector* collector : activeCollectors) {
    Message* m = collector->collect();
    if ( m->hasContents() ) { TempStorage::getInstance()->add(m); }
  }
}
```

This adds a new class `Message`. Also note that this adds a temporary storage between the packages `Collect` and `Transform` :



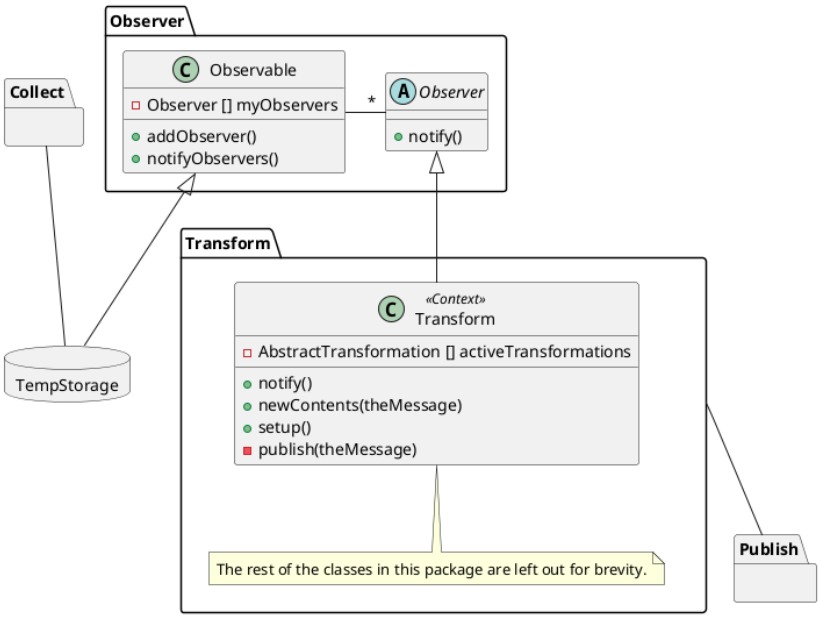
6.4 Package Model: The Transform Package

Implementing the Transformers using a `Strategy Pattern`.



6.5 Introducing an Observer Pattern

Just for the hell of it, so that the `TempStorage` can notify when there is new content added.



6.6 Package Model: The Publish Package

